

Does transformer paper-recommendation generalize across fields? A checkpointed, all-26-field test of SPECTER vs TF-IDF on held-out citation prediction

Author: Bucket Foundation · research-atlas working group **Version:** 2.0 (cross-field preprint draft) · **Date:** 2026-06-22
Corpus: OpenAlex, all 26 top-level fields, impact-ranked (most-cited first), 2015–2024 — checkpoint 1 = top ~3,000 works/field (78,000 works), grown by a checkpoint loop **DOI:** to be minted on the next research-atlas release (concept record on Zenodo; see `.zenodo.json`) **Reproducibility:** every number in this paper is emitted by `scripts/crossfield_run.py` → `analysis/crossfield/checkpoint_<N>.json` (+ `manifest.json` + `convergence.jsonl`), summarized by `scripts/crossfield_report.py`, and pinned by `tests/test_crossfield.py`. The checkpoint JSON is the authoritative source for every statistic quoted below; the paper reads from **checkpoint 1**.

Abstract

A companion single-subfield study showed that **SPECTER** (a transformer pre-trained on the scientific-paper citation graph) beats a TF-IDF baseline at held-out citation prediction in High-Energy Physics (+15.4% relative MAP, $p = 0.0005$). The natural question — and the one practitioners actually face when they reach for a neural paper-recommender — is whether that advantage **generalizes across fields**, or whether it is a property of one citation-dense subfield. We answer it directly. We built a **checkpointed, resumable, producer/consumer** pipeline that pulls an **impact-ranked corpus** (most-cited papers first) across **all 26 OpenAlex top-level fields**, builds a complete in-corpus citation graph and PageRank per field, embeds title+abstract on a local AMD GPU (ROCm) at a measured **9.1 docs/s steady-state**, and runs the identical held-out citation-prediction evaluation — SPECTER vs TF-IDF vs word2vec vs a text-free graph recommender, with bootstrap CIs and a paired test — in **every** field. At checkpoint 1 (top ~3,000 works/field, **78,000 works**), the verdict is nuanced and we report it honestly: **SPECTER beats TF-IDF on MAP in 16 of 26 fields**, but the across-field effect is **borderline** — combined field-level mean $\Delta\text{MAP} = +0.0095$ (95% CI $[-0.0005, +0.0195]$, one-sample bootstrap $p = 0.0624$); the sign test on 16/26 wins is not significant ($p = 0.33$). The advantage is real and large where text carries fine-grained meaning (Neuroscience +0.063 $p = 0.004$; Computer Science +0.060 $p < 0.001$; Social Sciences, Psychology), and **reverses** in several physical-science / pharmacology fields (Pharmacology -0.034 $p = 0.001$; Energy -0.014 $p = 0.001$; Earth Sciences -0.031 $p = 0.021$). Citation concentration varies by field (Gini **0.235–0.538**) and so does interdisciplinarity (cross-field reference fraction **0.114–0.468**). The headline: **transformer paper-recommendation does not robustly generalize across all fields — it is field-dependent, winning a majority but not all, with no significant aggregate edge**. The corpus grows by re-running the loop (the next tranche is top 5k, then 20k, then 50k works/field); every checkpoint is durable and the run resumes cleanly after interruption.

1. Introduction

The predecessor of this line of work is a course project (UMass MA544) that ranked and recommended arXiv `hep-ph` papers with TF-IDF+NMF topics, a cosine recommender, mean-pooled word2vec, and PageRank. A first paper (companion; `RESULTS.md`) rebuilt it on a **complete** OpenAlex citation graph for one subfield (Nuclear & High-Energy Physics), fixed the PageRank-uniformity artifact (Gini 0.53 instead of "differences at the 14th significant digit"), and added the quantitative evaluation it lacked: on 2,359 HEP queries, SPECTER beat TF-IDF on held-out citation prediction by **+15.4% relative MAP** with a paired-bootstrap $p = 0.0005$.

A single-subfield win is suggestive, not conclusive. SPECTER is trained on the citation graph; HEP is citation-dense and textually distinctive; both could flatter the transformer. The scientifically interesting — and practically decisive — question is **generalization**:

> Does a paper-trained transformer recommender beat a TF-IDF baseline at held-out > citation prediction **in every field**, or only in some? If only some, which, > and is there a net advantage across the literature as a whole?

This paper answers that question with a real, all-26-field measurement, and reports the answer whether or not it is the convenient one. The contribution is three-fold:

1. 1. **A method**: a checkpointed, resumable, ingestion-concurrent-with-analysis

orchestrator (`atlas/ranking/crossfield.py`) that scales the proven single-subfield pipeline to all 26 OpenAlex top-level fields, impact-ranked, growing the corpus in durable tranches.

1. 2. **A measurement**: the per-field held-out citation-prediction eval (SPECTER vs

TF-IDF vs word2vec vs graph) in all 26 fields at checkpoint 1, with bootstrap CIs and a paired test per field plus a combined field-level test.

1. 3. **An honest finding**: SPECTER wins a **majority** of fields but **not all**,

and the **aggregate** advantage is **not significant** — transformer paper-recommendation is **field-dependent**, not a universal win.

2. Data

2.1 Impact-ranked corpus across all 26 fields

For each OpenAlex top-level field (`primary_topic.field.id:fields/<id>`, articles, 2015–2024) we pull works **ordered by global cited_by_count descending** — the most impactful papers first — with each work's `referenced_works` (out-edges) and `cited_by_count` (global impact) and abstract. Impact-ranked ingestion is deliberate: it surfaces each field's canonical core before its long tail, so even a modest tranche is the part of the field that most recommendation traffic actually concerns. The corpus grows in **checkpoint tranches** (top 3k → 5k → 20k → 50k works/field, configurable); checkpoint 1 is the top ~3,000 most-cited works in every field.

metric (checkpoint 1)	value
fields	26
works (top ~3k/field)	78,000
publication window	2015–2024
GPU embedding model	SPECTER (<code>allenai-specter</code>), 768-d
GPU steady-state throughput	9.1 docs/s (AMD RX 7700S / ROCm)
wall-clock (checkpoint 1)	~52 min

(Source: `analysis/crossfield/checkpoint_1.json`.)

Raw OpenAlex pages are cached per (field, page-index); the per-id SPECTER embeddings are cached one `.npy` per work. Both caches make the run **resumable**: a re-run resumes from disk, and a larger tranche fetches only the new pages.

2.2 The honest coverage boundary, per field

As in the single-subfield study, the in-corpus citation graph is **complete by construction within each field's loaded slice** (no surviving edge dropped), while **global cited_by_count** carries impact from outside the slice. Because the corpus is impact-ranked and bounded per field, in-corpus edge density is lower than in the deep HEP slice — this is expected and is exactly why we (a) report it explicitly and (b) **grow** the corpus via the checkpoint loop: density rises with each tranche, and the convergence log tracks how the headline numbers move as it does.

3. Methods

3.1 The checkpointed, resumable, concurrent orchestrator

The engineering is part of the contribution, because a 26-field neural study on one GPU must survive interruption (we have lost connectivity mid-run before).

- - **Producer/consumer.** A network-bound **producer** thread downloads each

field's tranche (caching raw pages); the GPU/CPU-bound **consumer** analyzes a field the instant its raw is cached. The GPU never waits on the network — in checkpoint 1 the producer finished downloading all 26 fields while the consumer was still embedding the early ones.

- - **Durable checkpointing.** Each field's result is written to a per-field

partial file immediately on completion; when all fields finish, a single `checkpoint_<N>.json` is written atomically and the partial is dropped. A `manifest.json` indexes every checkpoint and a `convergence.jsonl` logs how per-field MAP / Gini / the SPECTER-vs-TF-IDF delta move as the corpus grows.

- - **Crash-safety / resume.** A SIGINT or crash mid-checkpoint loses nothing: the

per-field partial, the raw-page cache, and the per-id embedding cache are all on disk. Re-running resumes finished fields for free and re-analyzes only the rest; re-running after a clean checkpoint **advances** to the next tranche. A PID lock prevents two runs racing on the shared caches.

3.2 Per-field pipeline (identical to the proven single-subfield study)

For each field we reuse the companion study's modules unchanged:

- - **Complete in-corpus citation graph** (CSR) + **PageRank** (power method,

damping 0.85), with heavy-tail diagnostics (Gini, cv, top-1% mass).

- - **Four recommenders**, all scored on the identical closed candidate pool:

TF-IDF cosine (the baseline), **word2vec mean-pool cosine**, **SPECTER transformer cosine** (768-d, GPU), and a **text-free graph co-citation** recommender (bibliographic coupling).

- - **Held-out citation-prediction eval:** mask 30% of each eligible query's

in-corpus references, rank all candidates, score the held-out set with **Recall@k / MAP / MRR** and bootstrap 95% CIs; the SPECTER-vs-TF-IDF gap is tested with a **paired bootstrap** on per-query average precision. To keep the

graph dense and the candidate pool closed (so masked references are real in-pool targets), the eval runs on the citation-densest sample per field, exactly as in the single-subfield study.

3.3 Cross-field analyses

Beyond per-field tables we compute three cross-field quantities:

- - **Generalization.** In how many of 26 fields does SPECTER beat TF-IDF on MAP?

We report per-field Δ MAP and p, a **sign test** on the win count, and a **combined field-level test** — a one-sample bootstrap on the distribution of per-field MAP deltas (is the *across-field* mean delta > 0?).

- - **Concentration.** Citation-count and PageRank **Gini by field**, and their

ranges — how unequal impact is, field by field.

- - **Interdisciplinarity.** For each field, the fraction of its in-corpus

references whose target lives in a **different** top-level field (computed over the union of all 26 loaded fields, so cross-field edges are resolvable).

3.4 GPU embedding throughput

SPECTER runs on an AMD Radeon RX 7700S via ROCm (HSA_OVERRIDE_GFX_VERSION= 11.0.0 before importing torch; torch 2.9.1+rocm6.4), batch-encoding at a **measured steady-state of 9.1 docs/s** (excluding the one-time model load — the prior "~8/s" figure included it). On an 8 GB card SPECTER (a BERT-class, 512-token model) is genuinely encode-bound at batch 64; this is the honest rate, and it is why the neural study is checkpointed and resumable rather than run in one shot.

4. Results

4.1 The generalization verdict: a majority, not a universal, win

At checkpoint 1, across all 26 fields:

quantity	value	source
fields evaluated	26	crossfield.generalization.fields_evaluated
fields where SPECTER > TF-IDF (MAP)	16 / 26	fields_specter_wins
win fraction	0.62	win_fraction
sign test (16/26 wins)	p = 0.33	sign_test_p
combined field-level mean ΔMAP	+0.0095	combined_field_level.mean
combined 95% CI	[-0.0005, +0.0195]	combined_field_level.ci
combined bootstrap p	0.0624	combined_field_level.p

(Source: analysis/crossfield/checkpoint_1.json → crossfield.generalization.)

The honest reading: **SPECTER wins more fields than it loses (16 vs 10), but the aggregate advantage across fields is borderline and not significant** (combined CI just includes zero; sign test far from significant). A practitioner choosing a recommender for an arbitrary field cannot assume the transformer will win — it probably will, but with real risk of a tie or a loss, and the expected gain is small in the field-aggregate.

Macro-averaged over the 26 fields, the method ordering is nonetheless SPECTER-first:

method	macro-avg MAP (26 fields)
SPECTER transformer	0.196
TF-IDF cosine — <i>baseline</i>	0.187
word2vec mean-pool — <i>baseline</i>	0.158
graph co-citation (text-free)	0.089

(Macro-average of `methods.<m>.map` over fields. The transformer leads on average, but the lead is carried by a subset of fields — see §4.2.)

4.2 Where SPECTER wins, where it loses (per-field)

Sorted by ΔMAP (SPECTER – TF-IDF). Positive = SPECTER wins.

field	eval q	TF-IDF MAP	SPECTER MAP	ΔMAP	p	win
Neuroscience	136	—	—	+0.063	0.004	✓
Computer Science	528	—	—	+0.060	0.000	✓
Social Sciences	133	—	—	+0.039	0.015	✓
Psychology	133	—	—	+0.035	0.044	✓
Dentistry	162	—	—	+0.031	0.084	✓
Environmental Science	247	—	—	+0.030	0.013	✓
Nursing	74	—	—	+0.029	0.279	✓
Agricultural & Biological	150	—	—	+0.029	0.077	✓
Decision Sciences	128	—	—	+0.024	0.200	✓
Business, Management & Acc.	215	—	—	+0.023	0.097	✓
Materials Science	367	—	—	+0.020	0.049	✓
Engineering	376	—	—	+0.014	0.137	✓
Medicine	325	—	—	+0.014	0.207	✓
Economics, Econ. & Finance	101	—	—	+0.010	0.680	✓
Veterinary	134	—	—	+0.005	0.825	✓
Biochem., Genetics & Mol. Bio.	258	—	—	+0.000	0.980	✓
Physics and Astronomy	551	—	—	−0.001	0.860	✗
Chemistry	352	—	—	−0.004	0.759	✗
Chemical Engineering	683	—	—	−0.013	0.015	✗
Energy	769	—	—	−0.014	0.001	✗

Health Professions	60	—	—	−0.018	0.507	✗
Mathematics	207	—	—	−0.018	0.074	✗
Arts and Humanities	63	—	—	−0.020	0.493	✗
Immunology & Microbiology	102	—	—	−0.022	0.245	✗
Earth & Planetary Sciences	242	—	—	−0.031	0.021	✗
Pharmacology, Tox. & Pharm.	292	—	—	−0.034	0.001	✗

(Source: `checkpoint_1.json` → `fields[ ].transformer_vs_tfidf`; full per-method MAP/Recall/MRR tables with CIs are in the JSON and in `CROSSFIELD_RESULTS.md`.)*

A clear pattern emerges. SPECTER's biggest, most significant wins are in fields where **fine-grained phrase meaning carries the relevance signal** — Neuroscience, Computer Science, Social Sciences, Psychology. Its **significant losses** cluster in **physical-science and pharmacology** fields — Pharmacology, Energy, Earth & Planetary, Chemical Engineering — where TF-IDF's exact-term matching (compound names, instruments, materials, reaction terms) is hard to beat and SPECTER's semantic smoothing can *hurt*. Physics and Astronomy, the top-field that contains the single-subfield study's HEP slice, is a near-tie at this top-level granularity (ΔMAP −0.001), a useful reminder that a win in one citation-dense subfield need not survive aggregation to the whole field.

4.3 Concentration (Gini by field)

quantity	range across 26 fields
citation-count Gini	[0.235, 0.538]
PageRank Gini	[0.263, 0.652]

(Source: `crossfield.concentration`.)

Every field is heavy-tailed (no field is uniform — the single-subfield study's fix holds everywhere), but concentration varies: Decision Sciences and Computer Science are the most citation-concentrated (Gini $\approx$ 0.48–0.54), Pharmacology and Materials Science the least ($\approx$ 0.24–0.25). PageRank Gini runs higher and wider (up to 0.65), as recursive prestige amplifies the tail.

4.4 Interdisciplinarity

quantity	range across 26 fields	mean
fraction of references crossing field boundaries	[0.114, 0.468]	$\sim$ 0.30

(Source: `crossfield.interdisciplinarity`, computed over the union of all 26 loaded fields.)

The most inward-looking field is **Physics and Astronomy** (only $\sim$ 11% of its references leave the field), the most outward-looking are **Social Sciences**, **Health Professions**, and **Immunology & Microbiology** ($\sim$ 44–47% cross a field boundary). Interdisciplinarity does not cleanly predict the SPECTER win/loss split — Physics (low interdisc.) and Social Sciences (high interdisc.) sit on opposite sides of the verdict — which argues against a simple "neural wins where fields mix" story.

4.5 Convergence / coverage status

Checkpoint 1 is the **first** tranche (top $\sim$ 3k works/field). The corpus grows by re-running the loop; `convergence.jsonl` records, per checkpoint, the win count, the combined ΔMAP and its p, and the Gini range,

so a reader can watch whether the borderline aggregate result firms up (toward significance) or stays a wash as density increases. We make **no** claim that the checkpoint-1 numbers are converged; we claim they are the honest first measurement and that the method to refine them is built and running.

5. Discussion

Transformer paper-recommendation is field-dependent, not universal. This is the central finding and it is more useful than a blanket "transformers win." SPECTER's pre-training objective — pull a paper near the papers it cites — pays off most where the text *is* the relevance signal at fine grain (cognitive, computational, social-science abstracts) and can backfire where exact lexical tokens (chemical names, materials, pharmacological compounds, instruments) are the relevance signal and semantic smoothing blurs them. TF-IDF's "dumb" exact-match is a genuinely strong, sometimes stronger, baseline in those fields.

The single-subfield win did not cleanly survive generalization. HEP (a citation-dense subfield) gave SPECTER +15.4% MAP; the encompassing top-level field Physics and Astronomy is a tie at checkpoint 1. This is exactly the kind of result a generalization study exists to surface: an effect demonstrated in one favorable slice is not entitled to a cross-field claim, and ours doesn't get one.

The aggregate is borderline, and we let it be borderline. Combined mean $\Delta\text{MAP} +0.0095$ with a 95% CI of $[-0.0005, +0.0195]$ ($p = 0.0624$) is the textbook "almost, but not at $\alpha = 0.05$ " result. We resist the temptation to one-side the test or cherry-pick the 16 wins into a headline. The corpus loop exists precisely so this number can be re-measured at higher density rather than over-interpreted now.

Method choice should be field-aware. The practical recommendation that falls out of the table: use SPECTER (or a hybrid) in neuroscience/CS/social-science/ psychology; keep TF-IDF (or ensemble it ahead of the transformer) in pharmacology/energy/earth-science/chemistry. A field-agnostic "always neural" default is not supported by the data.

6. Limitations

Stated plainly; none hidden.

1. **Checkpoint 1 is one tranche.** Top ~3k works/field is the most-impactful core,

not the whole field. In-corpus edge density (and eval query counts, 60–769 per field) is lower than the deep HEP slice. The result is a first measurement that the loop is built to refine; the convergence log is the place to read whether it firms up.

1. 2. **Per-field power varies.** Fields with few dense queries (Health Professions

60, Arts & Humanities 63, Nursing 74) have wide per-field CIs; their individual p-values are weak. The **combined** field-level test is the right unit for the generalization claim, and it is borderline.

1. 3. **Top-level fields blur subfields.** A win in one subfield (HEP) can vanish when

averaged over its top-level field (Physics & Astronomy). The 26-field grain answers "does it generalize across fields"; it does not resolve subfield structure (a future checkpoint could descend to subfields).

1. 4. **Closed-world, citation-density-selected eval.** As in the single-subfield

study, each field's eval runs on its citation-densest sample so the graph isn't sparsified; absolute Recall/MAP/MRR are relative, not open-world retrieval quality.

1. 5. **word2vec baseline is in-domain PPMI-SVD**, not Google's binary — the representation class (static word vectors, mean-pooled, cosine) is identical; only the training corpus differs.

1. 6. **Citation prediction is a proxy for relevance** — large, objective, and annotation-free, but a proxy; methods that mimic citation habits (the graph recommender, strong at deep recall) are advantaged by it.

1. 7. **GPU-bounded**. All embeddings are from one 8 GB AMD GPU at ~9 docs/s; the neural study is therefore checkpointed. The same code scales with more GPU time — that is the loop.

7. Reproducibility statement

Every number is computed by `scripts/crossfield_run.py` and written to `analysis/crossfield/checkpoint_<N>.json` (+ `manifest.json` + `convergence.jsonl`), summarized by `scripts/crossfield_report.py` into `CROSSFIELD_RESULTS.md`, and pinned by tests that fail if prose and data diverge (`tests/test_crossfield.py`, including a guard asserting the cross-field headline against `checkpoint_1.json`, and tests for the cross-field aggregation + the producer/consumer resume logic).

```
pip install -e . # numpy, pandas, pyarrow, scikit-learn,
sentence-transformers, torch(+rocm)
# run / advance the loop (first run = checkpoint 1, top ~3k works/field):
HSA_OVERRIDE_GFX_VERSION=11.0.0 python scripts/crossfield_run.py
# re-run to GROW the corpus to the next tranche (5k -> 20k -> 50k works/field):
HSA_OVERRIDE_GFX_VERSION=11.0.0 python scripts/crossfield_run.py
# measure GPU embed throughput on the current cache:
HSA_OVERRIDE_GFX_VERSION=11.0.0 python scripts/crossfield_run.py --measure-throughput
# summarize the latest checkpoint into the per-field results doc:
python scripts/crossfield_report.py
python -m pytest tests/test_crossfield.py -q
```

The run is **resumable**: a SIGINT/crash mid-checkpoint loses nothing (per-field partial + raw-page cache + per-id embedding cache), and re-running continues from the last checkpoint.

Data availability. The heavy raw OpenAlex page cache and the per-id SPECTER embedding `.npy` cache are rebuilt from the OpenAlex API and are gitignored; the checkpoint JSONs, manifest, convergence log, and per-field results doc are committed. License: MIT (code) / CC-BY-4.0 (data, via OpenAlex CC0).

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Appendix A. Headline numbers (machine-checked, checkpoint 1)

Quantity	Value	Source field
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Fields evaluated	26	fields_with_results
Works (top ~3k/field)	78,000	total_works_loaded
GPU embed throughput (steady-state)	9.1 docs/s	embed_docs_per_sec
SPECTER > TF-IDF (MAP)	16 / 26 fields	crossfield.generalization.fields_specter_wins
Win fraction	0.62	win_fraction
Sign test p	0.33	sign_test_p
Combined field-level mean Δ MAP	+0.0095	combined_field_level.mean
Combined 95% CI	[-0.0005, +0.0195]	combined_field_level.ci
Combined bootstrap p	0.0624	combined_field_level.p
Citation Gini range	[0.235, 0.538]	concentration.citation_gini_range
PageRank Gini range	[0.263, 0.652]	concentration.pagerank_gini_range
Interdisciplinarity range	[0.114, 0.468]	interdisciplinarity.range
Embedding model	SPECTER (allenai-specter), 768-d, GPU/ROCm	transformer_model